

**To Receive Full Credit, You Must Show All Your Work**

The LeBlanc process produces "black ash" which is a solid consisting of 42 wt% sodium carbonate (C), 6 wt% water-soluble material (S), and 52 wt% water-insoluble material (I). The black ash is treated with liquid water (W) to extract the sodium carbonate. The result of the process produces a solid residue containing: 4 wt% C, 0.5 wt% S, 85 wt% I, and 10.5 wt% W. Also, a liquid product containing C, S, and W is produced. Use a basis of  $m_1 = 1$  ton of black ash. Label the mass of liquid water added as  $m_2$ , the mass of the solid residue produced as  $m_3$ , and the mass of the liquid product as  $m_4$ . Give all your answers to the tenth of a lb<sub>m</sub>.

- (a) Draw and label a block flow diagram. Define a system.  
 (b) Calculate the mass of the solid residue,  $m_3$ , the mass of the sodium carbonate extracted to the liquid product,  $m_{4,C}$ , and the mass of S extracted to the liquid product,  $m_{4,S}$ , in lb<sub>m</sub>.  
 (c) If the extraction process requires that the liquid product to be 75 wt% water, what is the mass in lb<sub>m</sub> of the required liquid water added to the black ash, that is,  $m_2$ , and the mass of the liquid product,  $m_4$ ?

C = sodium carbonate  
 S = water soluble mat  
 I = water insoluble  
 W = Water

$m_1 = 1$  ton black ash

$m_2 = \text{mass W}$

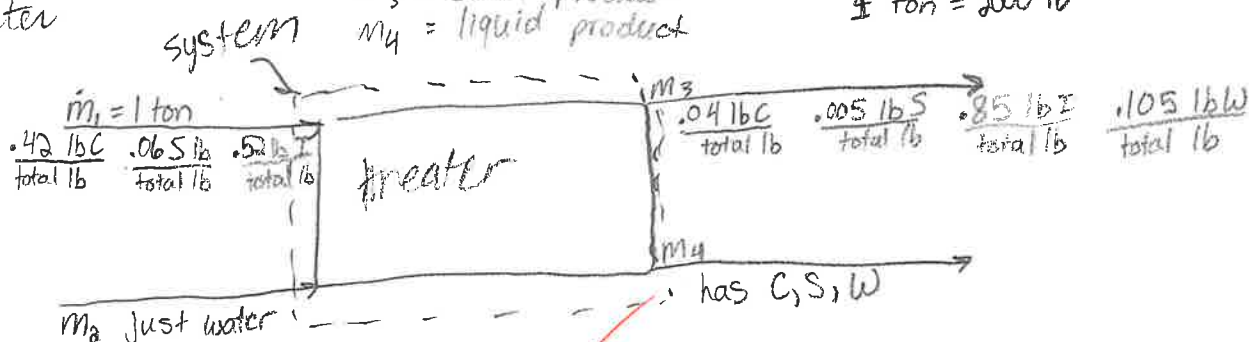
$m_3 = \text{solid product}$

$m_4 = \text{liquid product}$

1 lb<sub>m</sub>

1 lb =  $5 \times 10^{-4}$  ton

1 ton = 2000 lb



Mass balance:  $\dot{m}_1 + \dot{m}_2 = \dot{m}_3 + \dot{m}_4$  ✓

I balance:  $\dot{m}_1 X_{1,I} = \dot{m}_3 X_{3,I}$

$2000 \text{ lb total} \times \frac{.52 \text{ lb I}}{\text{total lb}} = \dot{m}_3 \times \frac{.85 \text{ lb I}}{\text{total lb}}$

$1040 \text{ lb I} \times \frac{\text{total lb}}{.85 \text{ lb I}} = 1223.5 \text{ lb total} = \dot{m}_3$  ✓

S Balance:  $\dot{m}_1 X_{1,S} = \dot{m}_3 X_{3,S} + \dot{m}_4 X_{4,S}$

$2000 \text{ lb tot} \times \frac{.06 \text{ lb S}}{\text{tot lb}} = 1223.5 \text{ lb tot} \times \frac{.005 \text{ lb S}}{\text{lb tot}} + \dot{m}_4 X_{4,S}$

$120 \text{ lb S} - 6.1175 = 123.9 \text{ lb in } \dot{m}_4 X_{4,S}$  ✓

C Balance:  $\dot{m}_1 X_{1,C} = \dot{m}_3 X_{3,C} + \dot{m}_4 X_{4,C}$

$2000 \text{ lb tot} \times \frac{.42 \text{ lb C}}{\text{total lb}} = \frac{.04 \text{ lb C}}{\text{total lb}} \times 1223.5 + \dot{m}_4 X_{4,C}$

$835.1 \text{ lb C} = \dot{m}_4 X_{4,C}$

$799.1 \text{ lb C}$

calc error

W lb

$123.9 \text{ lb S} + 835.1 \text{ lb C} + \text{lb W} = .75$

$\text{W lb} = 719.22 \text{ lb} + .75 \text{ lb W}$

$.25 \text{ W lb} = 719.22 \text{ lb}$

$\text{W} = 2876.8 \text{ lb in } \dot{m}_2$  ✓

Water in  $M_4$   
 $2000 \text{ lb} + 2876.8 \text{ lb} =$   
 $1223.5 + \dot{m}_4$   
 $\dot{m}_4 = 3653.3 \text{ lb}$

OK

A bit of a mess, but you seem to have the right idea.

